Lesson 2 Teacher's Key

001 Game Creator (Windows only)

INSTRUCTOR ONLY - do not distribute to students

Level 1 - Complete Worked Solution

One Input Trigger:

```
Input Trigger: Right Arrow (Pressed) -> Position Actor(main, X: main.X +
8, Y: main.Y)
```

Level 2 - Complete Worked Solution

Four Input Triggers, each a Control Combination bound to one arrow key set to "Pressed," each running a script with a single Position Actor event targeting main:

```
Input Trigger: Right Arrow (Pressed) -> Position Actor(main, X: main.X +
8, Y: main.Y)
Input Trigger: Left  Arrow (Pressed) -> Position Actor(main, X: main.X -
8, Y: main.Y)
Input Trigger: Up    Arrow (Pressed) -> Position Actor(main, X: main.X,
Y: main.Y - 8)
Input Trigger: Down  Arrow (Pressed) -> Position Actor(main, X: main.X,
Y: main.Y + 8)
```

Level 3 - All Worked Solutions

Every stretch-goal idea from the handout, fully worked - not just one. Useful for validating each idea actually works before class, and as a ready reference no matter which one a student ends up picking (or if a student blows past all of them, this is your own jumping-off point for exploring further right alongside them).

Animate: blink while moving, static when idle

001 Game Creator has no press-and-hold movement without extra work (see the Super-Stretch Ideas list), so walking here means repeatedly pressing and releasing a direction key - and every release genuinely passes through a real moment with no key held, even mid-stride. That means the idle-reset script built for the Recolor goal below already produces this goal too, for free: main's sprite visibly swaps between its direction sprite and its idle sprite on every step, which reads as a blink while walking and goes static the instant a student actually stops moving. No separate event needed - see Recolor for the scripts. Worth calling out to the class directly as a deliberate consequence, not a coincidence: this is what happens when idle-detection is built from key state under a press-and-release movement scheme, and it will disappear on its own once movement switches to true press-and-hold (Change Movement X/Y, a Super-Stretch idea), since Released will then only fire on a genuine stop.

Sound: play on movement, silent otherwise

Import sfx_blip.wav (Resources > Import > Sound), then add a "Play Sound" event directly after each of the 4 Position Actor events. Because each Input Trigger fires on "Pressed" rather than continuously, this is naturally a one-shot per key press - no cooldown timer needed, unlike engines that poll movement every frame.

Recolor: different sprite per direction, reset when idle

Pair each of the 4 Position Actor events with a Change Body/Clothing Sprite event for that direction - swapping main's whole Sprite for a different pre-made colored one, rather than tinting the sprite it already has. Import the other placeholder circles as extra Sprites (Resources > Sprites > Create New > No Template, same process as main's original appearance in Setup) so you have one Sprite per color to swap between. 001 Game Creator's Input Triggers don't have a standing "nothing is pressed" trigger, so build idle detection instead: add a second Control Combination for each arrow key, this one set to "Released" rather than "Pressed," and give it a Multi-Control Branch checking all 4 arrow keys - the last path Multi-Control Branch provides is "none of these are pressed," which is the actual idle moment, so that's where the reset-to-default Change Body/Clothing Sprite event goes. Checking all 4 keys (not just "this key released") matters because a student could release Right while still holding Up - the sprite should only reset once every direction is let go. These same two scripts (per-direction Pressed sprite-swap + shared idle Released reset) are also what produces the Animate goal above, as a side effect - see that entry for why. Same Idle/Moving concept as the State Diagram from this lesson's bonus section.

Clamp: automatic, no scripting needed

This is a genuine freebie in 001 Game Creator, not something to build - open the Map's Properties and look at the Boundaries section (bottom-right of the window): a checkmark for each of the 4 directions plus Floor and Ceiling, all ticked by default, and a Connection dropdown per edge for linking to another Map. With every direction ticked and every Connection left at (None), the engine itself refuses to let main cross an edge - Position Actor moves that would push main past the boundary are simply stopped before they happen. That's why a "Crossed Map's Edge" Actor trigger doesn't behave like a live "player tried to leave" hook here: with the boundary actively blocking movement, main's position never actually reaches past the edge during ordinary play, so there's nothing for that trigger to catch beyond a one-time check when the Actor is first placed. It's worth understanding this rather than just accepting it, because it's easy to accidentally break: unticking one of those 4 boundary checkmarks, or setting a Connection to a second Map, lets main walk straight off that edge - the opposite of the Clamp goal - so treat those Boundaries settings as something to leave alone unless you deliberately want to add a second Map later.

Structure: group everything triggered by the same action in one place

Each direction's own Input Trigger script already holds that direction's movement, sound, and recolor (which also drives that direction's share of the Animate blink), with the shared idle-reset living in the Released scripts rather than duplicated per direction. Clamp needs no script anywhere - it's a Map Properties setting, not an event - so the Complete Level 3 solution below only has the 8 Input Trigger scripts (4 Pressed, 4 Released) plus a note about where Clamp actually lives.

Complete Level 3 solution (all 5 goals combined)

Everything above combined - one script per direction, plus one boundary script:

```
Input Trigger: Right Arrow (Pressed) ->
  Position Actor(main, X: main.X + 8, Y: main.Y)
  Play Sound(sfx_blip.wav)
  Change Body/Clothing Sprite(main, Sprite: player_circle_red)

Input Trigger: Left Arrow (Pressed) ->
  Position Actor(main, X: main.X - 8, Y: main.Y)
  Play Sound(sfx_blip.wav)
  Change Body/Clothing Sprite(main, Sprite: player_circle_blue)

Input Trigger: Up Arrow (Pressed) ->
  Position Actor(main, X: main.X, Y: main.Y - 8)
  Play Sound(sfx_blip.wav)
  Change Body/Clothing Sprite(main, Sprite: player_circle_green)

Input Trigger: Down Arrow (Pressed) ->
  Position Actor(main, X: main.X, Y: main.Y + 8)
  Play Sound(sfx_blip.wav)
  Change Body/Clothing Sprite(main, Sprite: player_circle_yellow)

Input Trigger: Right/Left/Up/Down Arrow (Released - one Control
Combination per key, in addition to each key's Pressed combination above)
->
  Multi-Control Branch(Right Arrow, Left Arrow, Up Arrow, Down Arrow)
    any path where a key is still pressed -> (no action - stay in that
direction's sprite)
    "none pressed" path -> Change Body/Clothing Sprite(main, Sprite:
player_circle_orange)

(This same Released/Multi-Control Branch pair is also what makes main
blink while walking - see the Animate idea above for why. No separate
animation event needed.)

(Clamp needs no script at all - Map Properties > Boundaries is ticked on
all 4 sides by default with no Connections set, so 001 Game Creator
already refuses to let main cross the edge. See the Clamp idea above.)
```

Alternate Path Notes: Textual Scripting

Reiterate to students in person: this mode switch is genuinely irreversible per-project. Confirm they created a separate throwaway project before letting them proceed.

Exact text-command names should be checked against the installed version's Script Editor autocomplete or the wiki before class, since this course otherwise treats 001 Game Creator as visual-only and hasn't needed to track Textual syntax changes.

- This path exists for students who've realized they prefer typing code (maybe from Godot or GameMaker) and want to see whether 001 Game Creator supports that style too - it does, with this one important caveat.
- Because of the irreversibility, this should stay an optional, teacher-supervised exploration rather than a path any student defaults into.